AP PRECALCULUS

Unit 1 • Lesson 1.1 notes

Lesson 1.1 • Change in Tandem

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Describe how the output values of a function change as the input values increase.
- Identify where a function is increasing or decreasing, and where its graph is concave up or concave down.
- Identify zeros of a function from a graph, a table or a formula.

Vocabulary. function, input, output, increasing, decreasing, rate of change, concave up, concave down, zero, x-intercept.

1. Inputs and outputs change together

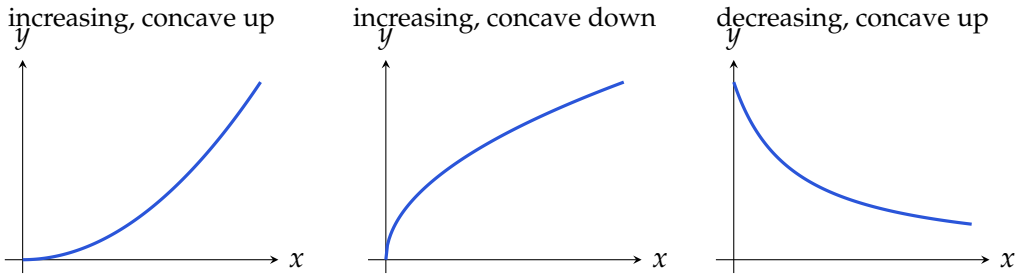
A function f pairs each input x with exactly one output $f(x)$. Reading a function means asking: *as x increases, what happens to $f(x)$?* The answer is the idea of **change in tandem**.

Increasing and decreasing

On an interval, f is **increasing** if $x_1 < x_2$ always gives $f(x_1) < f(x_2)$: the graph goes up from left to right. f is **decreasing** if $x_1 < x_2$ always gives $f(x_1) > f(x_2)$: the graph goes down.

2. How fast? Concavity

Two functions can both increase but in different ways. If the outputs increase by larger and larger amounts, the rate of change is increasing and the graph bends upward: it is **concave up**. If they increase by smaller and smaller amounts, the rate of change is decreasing and the graph is **concave down**. The same language applies to decreasing functions.



3. Zeros

A **zero** of f is an input a with $f(a) = 0$. On the graph, zeros are the x-intercepts. Between two consecutive zeros, a function that has no breaks keeps the same sign.

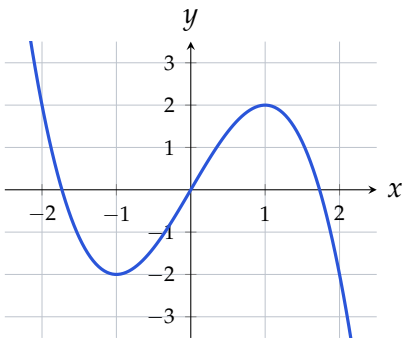
Example 1 • Reading a table

x	0	1	2	3	4	5
$g(x)$	-6	-1	2	3	2	-1

Solution. The outputs increase from $x = 0$ to $x = 3$ and decrease from $x = 3$ to $x = 5$. The changes are $5, 3, 1, -1, -3$: they decrease steadily, so the graph is concave down. The sign

changes between $x = 1$ and $x = 2$ and between $x = 4$ and $x = 5$, so (if g has no breaks) g has a zero in each of these intervals.

Example 2 • Reading a graph



The graph of $f(x) = -x^3 + 3x$ is shown.

Solution. f is decreasing on $(-\infty, -1)$, increasing on $(-1, 1)$, decreasing on $(1, \infty)$. The graph is concave up for $x < 0$ and concave down for $x > 0$. The zeros are $x = 0$ and $x = \pm\sqrt{3} \approx \pm 1.73$.

Common mistakes

- Describing intervals with y-values. Intervals of increase and decrease are given with *input* values (x).
- Confusing "decreasing" with "concave down". A graph can be increasing and concave down at the same time.

AP exam tip

When you justify concavity from a table, compare the changes over *equal-length* consecutive intervals and say whether they increase or decrease.

Quick check

1. Outputs at $x = 0, 1, 2, 3$ are $10, 7, 5, 4$. Is the function increasing or decreasing? Concave up or down?
2. Where is $f(x) = (x - 2)(x + 5)$ equal to zero?

Answers. 1. Decreasing (outputs fall); changes $-3, -2, -1$ are increasing, so concave up. 2. $x = 2$ and $x = -5$.